

BETHEL ASSEFA

Final-Year Mechanical Engineering Student

 **Addis Ababa, Ethiopia**

 **bethelassefa815@gmail.com**

 **+251 951 594 951**

Professional Summary

Final-year Mechanical Engineering student at Addis Ababa University with hands-on experience in renewable energy systems, CAD modeling, computational fluid dynamics (CFD), additive manufacturing, and composite fabrication. Experienced in designing and prototyping small-scale wind turbine blades using SolidWorks, ANSYS Fluent, QBlade, and Ultimaker Cura. Passionate about engineering design, sustainable energy, and solving real-world technical challenges through innovation and teamwork.

Education

Addis Ababa University

Bachelor of Science in Mechanical Engineering

Final-Year Student

Expected Graduation: *(We'll add the correct month and year.)*

Internship Experience

Mechanical Engineering Intern

CTBE Thematic Research

March 2026 – Present

- Designed and optimized a small-scale NACA 4424 wind turbine blade using Blade Element Momentum Theory (BEMT).
 - Performed aerodynamic CFD simulations using ANSYS Fluent to evaluate blade performance.
 - Designed split molds for composite blade manufacturing using SolidWorks.
 - Optimized 3D printing parameters in Ultimaker Cura to reduce material consumption and printing time.
 - Manufactured composite wind turbine blades using the hand lay-up process with fiberglass, epoxy resin, gel coat, and release wax.
 - Assisted in prototype development, testing, documentation, and technical reporting.
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Engineering Projects

Small-Scale Wind Turbine Blade Design and Prototyping

Tools: SolidWorks • ANSYS Fluent • QBlade • Ultimaker Cura

- Designed an aerodynamic wind turbine blade based on the NACA 4424 airfoil.
 - Conducted CFD simulations for aerodynamic validation.
 - Designed and manufactured 3D-printed molds.
 - Fabricated composite blades using hand lay-up techniques.
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Impact Testing Machine

- Designed and built a pendulum impact testing machine.
 - Integrated Arduino Uno and IR sensors.
 - Collaborated in fabrication, assembly, and testing.
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Four-Jaw Chuck Design

- Designed all components using SolidWorks.
- Produced engineering drawings and complete assembly.
- Created an assembly animation.

Automated Hand Sanitizer Dispenser

- Participated in the design and development of a touchless sanitizer dispenser.
 - Worked as part of a multidisciplinary engineering team.
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Technical Skills

CAD And Design

- SolidWorks

Simulation

- ANSYS Fluent
- MATLAB
- QBlade

Manufacturing

- 3D Printing
- Composite Manufacturing
- Hand Lay-up
- Mold Design

Engineering

- CFD Analysis
 - Technical Documentation
 - Mechanical Design
 - Problem Solving
 - Team Collaboration
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Additional Experience And Certifications

- Research And Development Assistance Virtual Internship Certificate
Benetsa Marketing and Communication | October 2025 – January 2026
 - Fundamentals of Digital Marketing
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Leadership And Activities

- Ambassador, AI and Robotics Club, Addis Ababa University
 - Member, AI and Robotics Club
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Languages

- Amharic — Native
 - English — Professional Working Proficiency
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